Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – II

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AIM: Working with Data Sources and Data Preparation in Tableau.

1. **Connect to different data sources (Excel, CSV, Database)**
2. **Create and use data extracts vs live connections**
3. **Manage metadata (rename, hide, change data types)**
4. **Perform joins (inner, left, right) on multiple tables**
5. **Apply data blending between two sources**
6. **Implement different types of filters (dimension, measure, date)**
7. **Optimize performance using extracts and filtering**

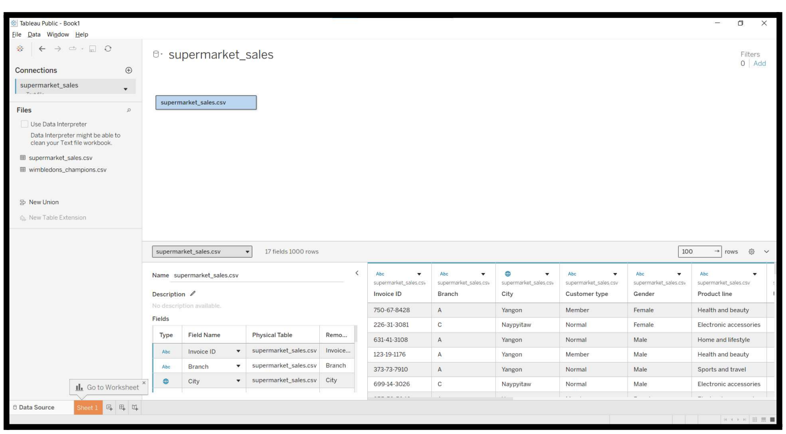
STEPS:

A. Connecting to Different Data Sources

Step 1 – Connect to the .CSV File

1. Open Tableau Desktop.
2. Under the Connect section, click on the text file.
3. Browse to the downloaded file superarket_sales.csv.
4. Select the file and click Open.
5. Wait for Tableau to load the dataset into the Data Source page.

Screenshot 1: Data Source page after loading supermarket_sales.csv.



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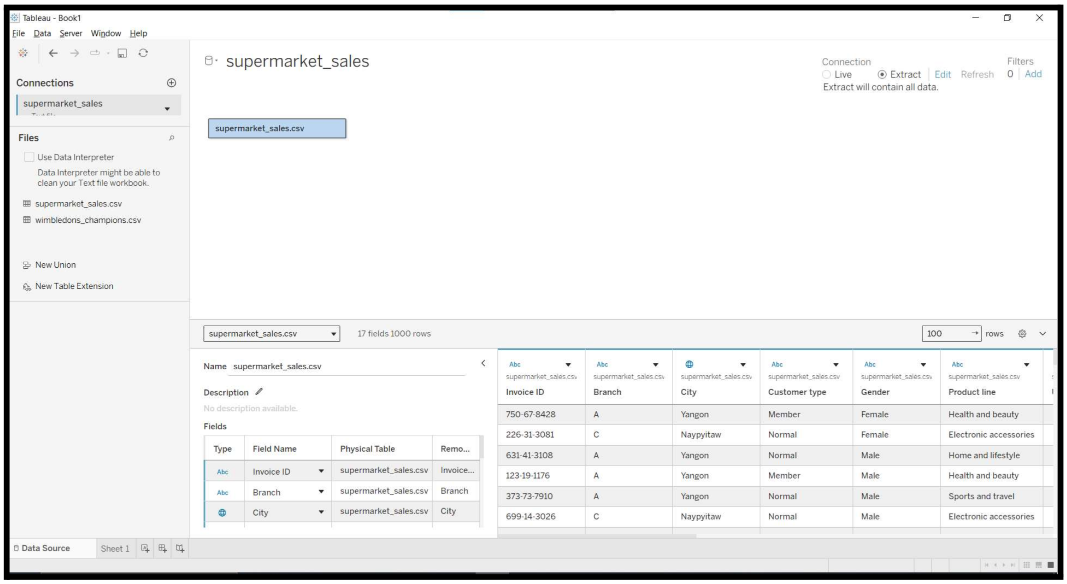
B. Creating Data Extracts vs. Live Connections

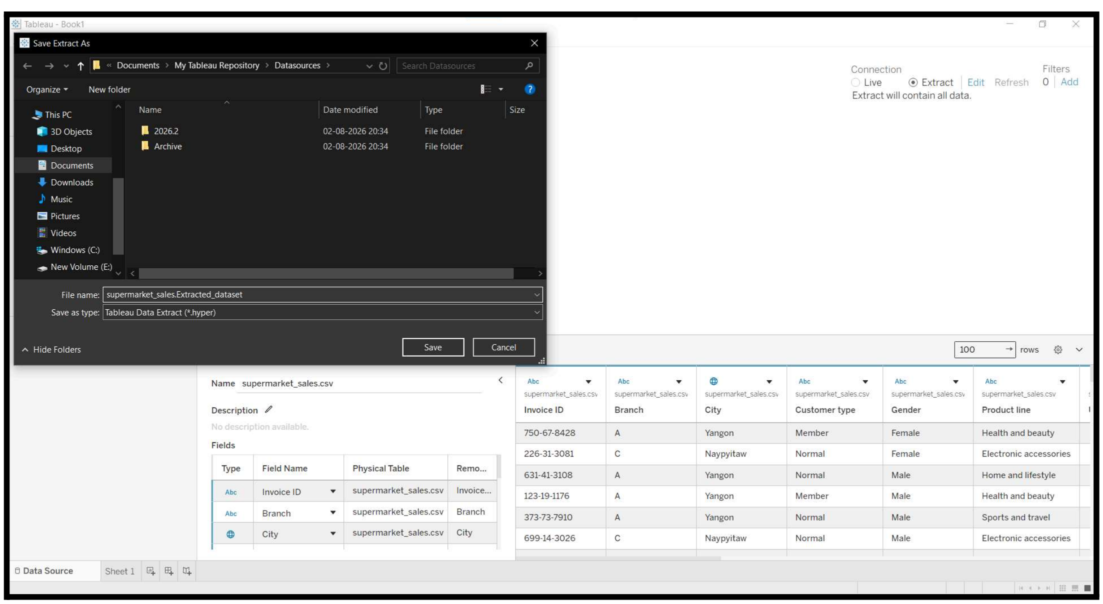
Step 2 – Create an Extract Connection

1. After the dataset is loaded in the **Data Source** tab, navigate to the **top-right corner** of the screen.
2. In the **Connection** section, switch the connection mode from **Live** to **Extract**.
3. Click the **Sheet 1** tab located at the **bottom-left** of the Tableau window.
4. When the **Save Extract** dialog box appears, enter **Ecommerce_Extract** as the file name.
5. Click **Save** to create and store the extract file, then continue working in the worksheet.

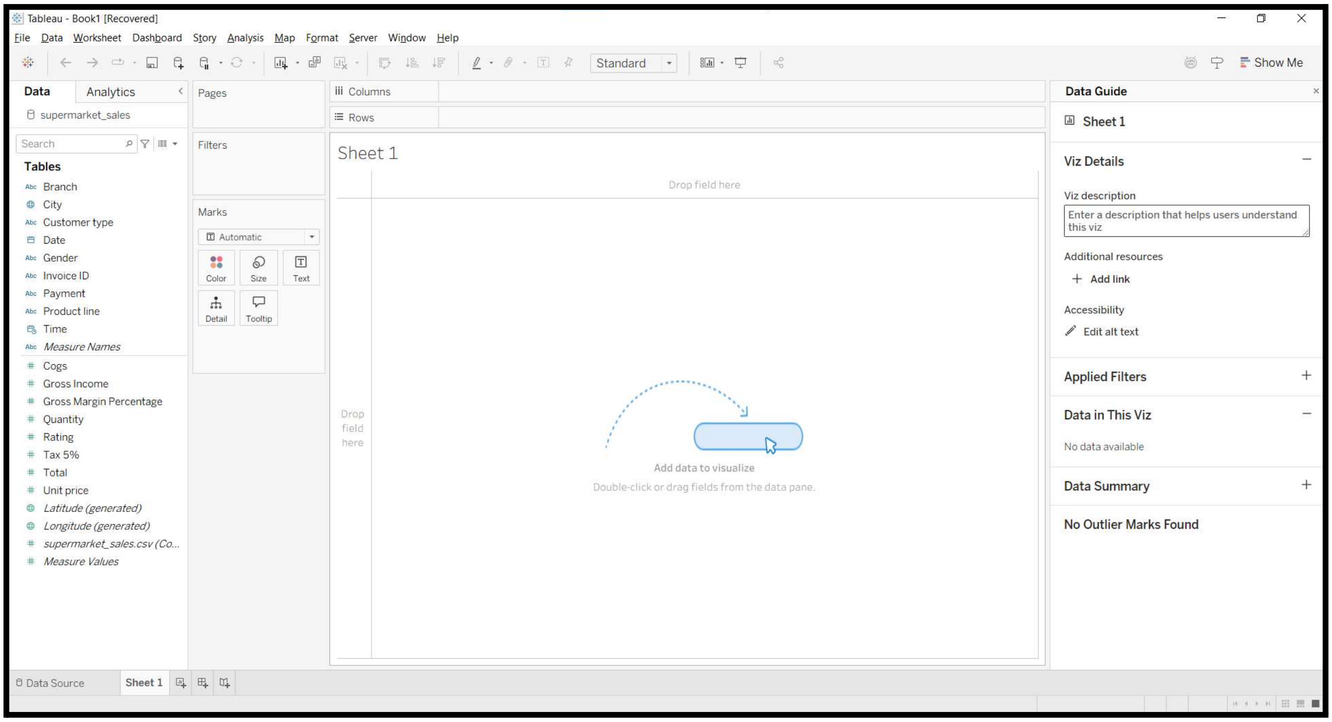
Screenshot 2: Connection mode changed from Live to Extract on the Data Source page.



Screenshot 3: Save Extract dialog box with the file name Ecommerce_Extract entered before saving.



Screenshot 4: Worksheet (Sheet 1) opened successfully after saving the extract file.

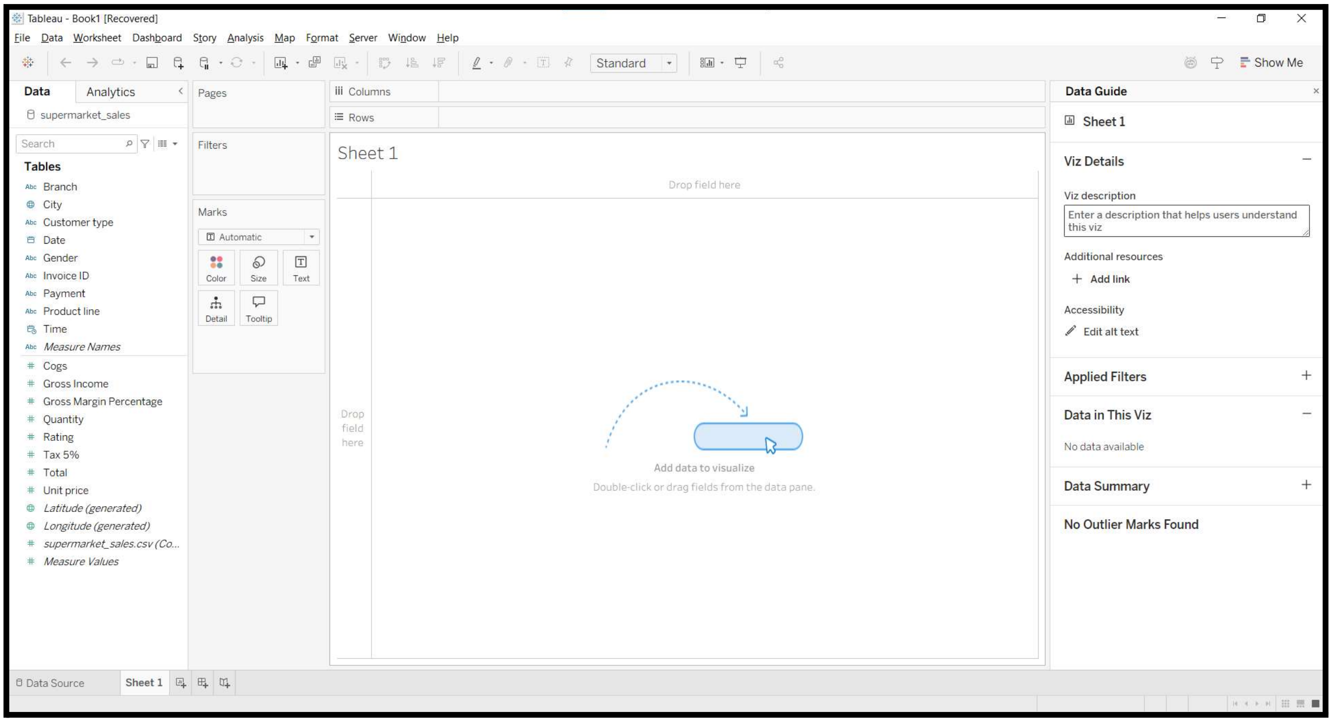


C. Managing Operational Metadata.

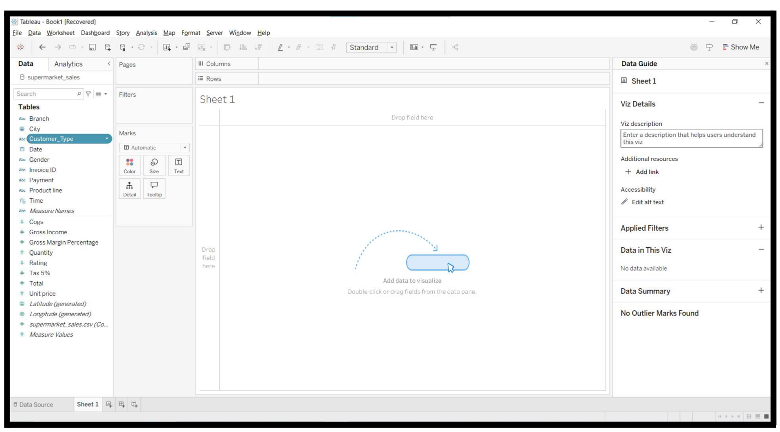
Step 3 – Rename, Hide, and Change Data Type of Fields

1. Open Sheet 1 and view the Data Pane on the left side of the workspace.
2. Locate the field Customer type, right-click it, select Rename, enter Customer_Type, and press Enter.
3. Locate the field Invoice ID, right-click it, and select Hide to remove it from the Data Pane.
4. Find the Date field and click the data type icon beside it.
5. If the field is detected as Text (Abc), change it to Date from the dropdown menu to ensure proper date recognition.

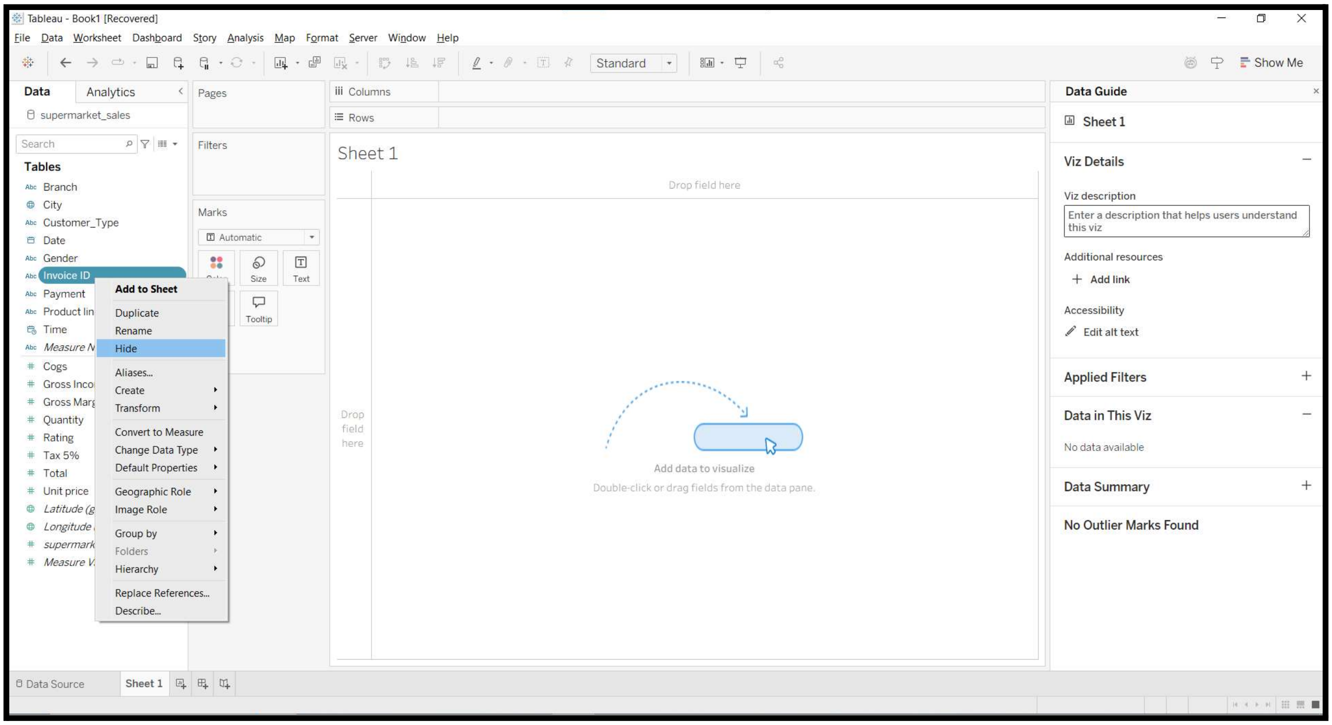
Screenshot 5: ***Data Pane in Sheet 1 displaying all fields from the supermarket_sales.csv dataset.***



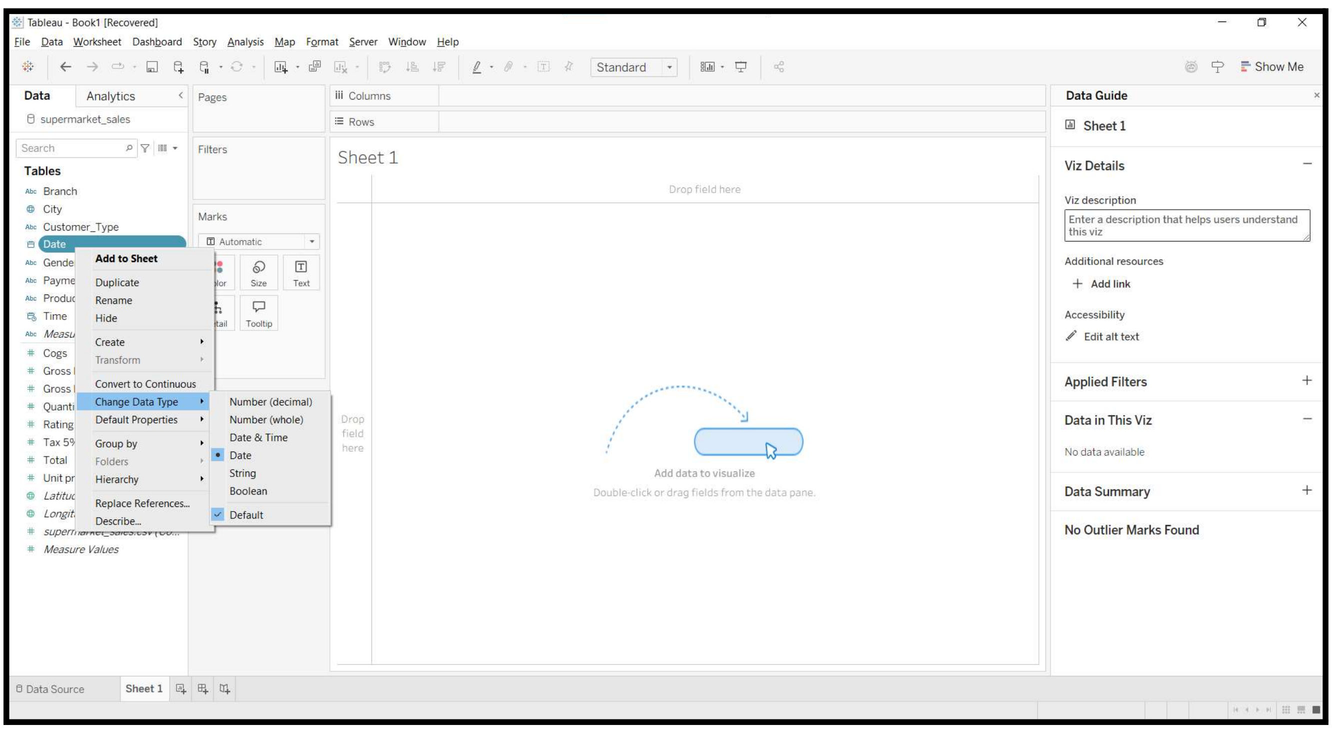
Screenshot 6: ***The Customer type field being renamed to Customer_Type.***



Screenshot 7: ***The Invoice ID field hidden from the Data Pane.***



Screenshot 8: ***The Date field data type changed to Date for correct date handling in Tableau.***

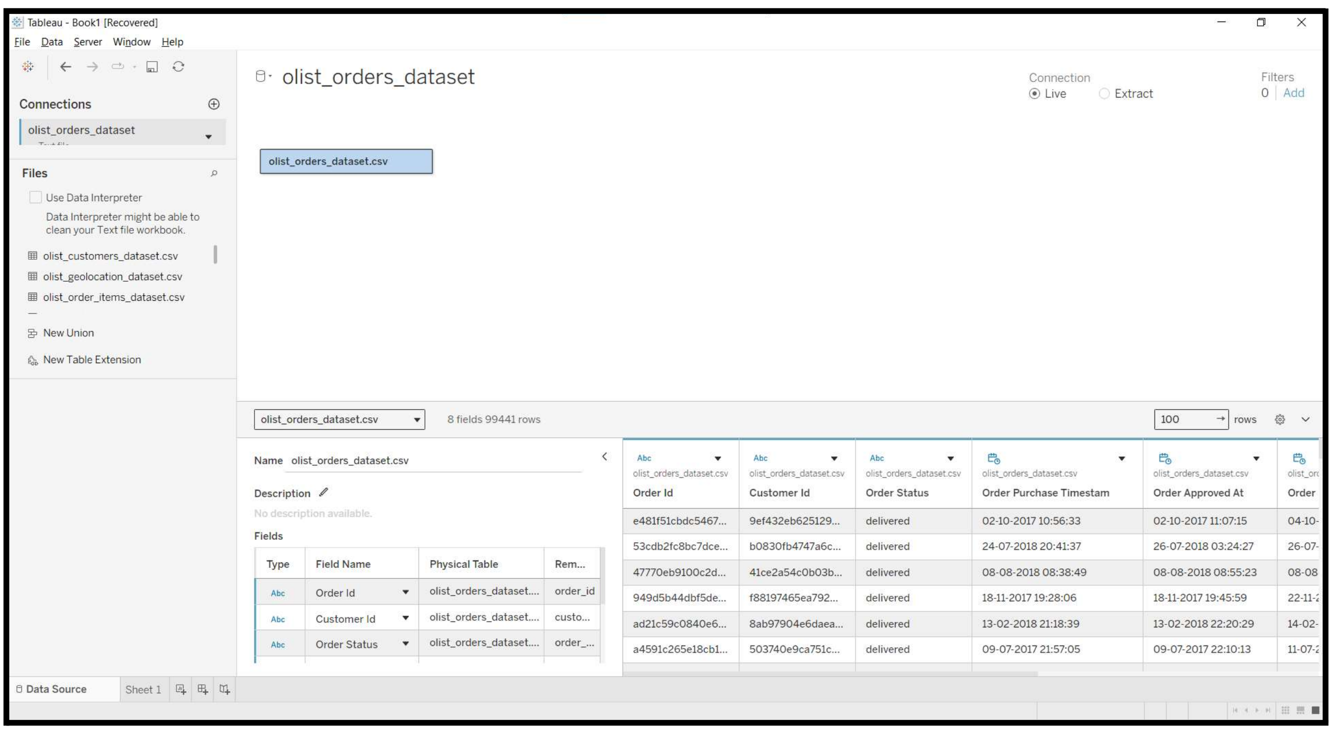


D. Performing Relational Joins on Multiple Tables.

Step 4 – Join Orders and Order Payments Tables

1. Click the **Data Source** tab at the bottom-left corner to return to the Data Source page.
2. Locate the **olist_orders_dataset.csv** table on the canvas.
3. Double-click the **olist_orders_dataset.csv** table to enter the **Physical Layer**.
4. From the left pane, drag **olist_order_payments_dataset.csv** next to the **olist_orders_dataset.csv** table to create a join.
5. Click the **Join** (Venn Diagram) icon between the two tables.
6. Select **order_id** as the matching field for both the left and right tables. (Tableau may select it automatically.)
7. Test the join types:
    a. Select **Inner Join** to display only matching order and payment records.
    b. Select **Left Join** to display all orders and their matching payment details, leaving unmatched payment fields blank.
8. Keep the join type set to **Left Join** and continue with the analysis.

Screenshot 9: ***Data Source page showing the olist_orders_dataset.csv table loaded.***



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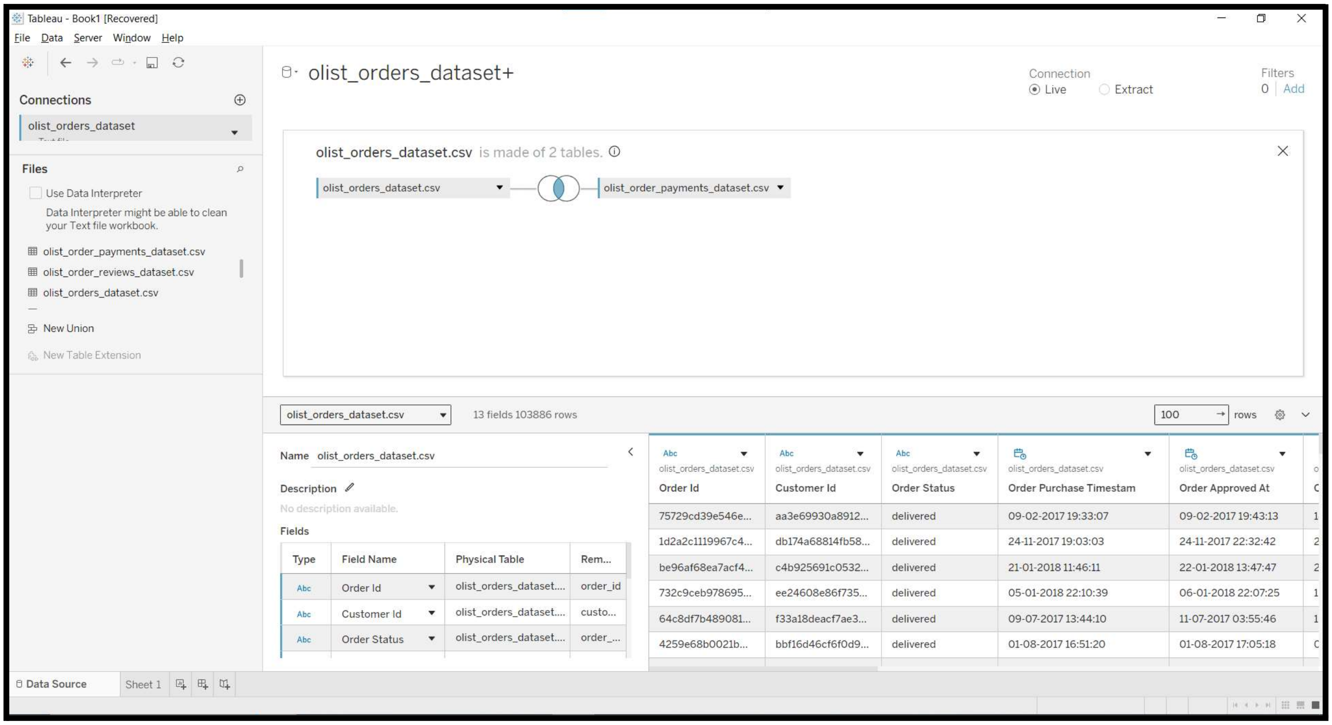
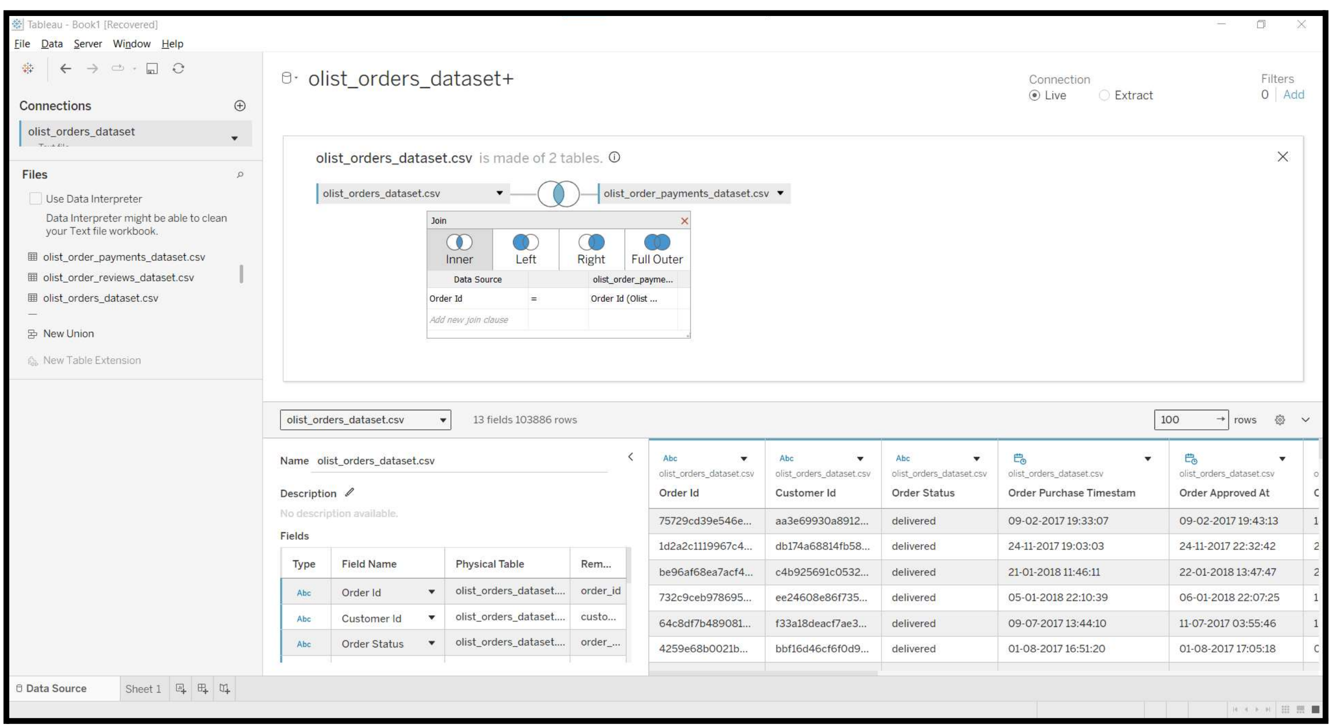
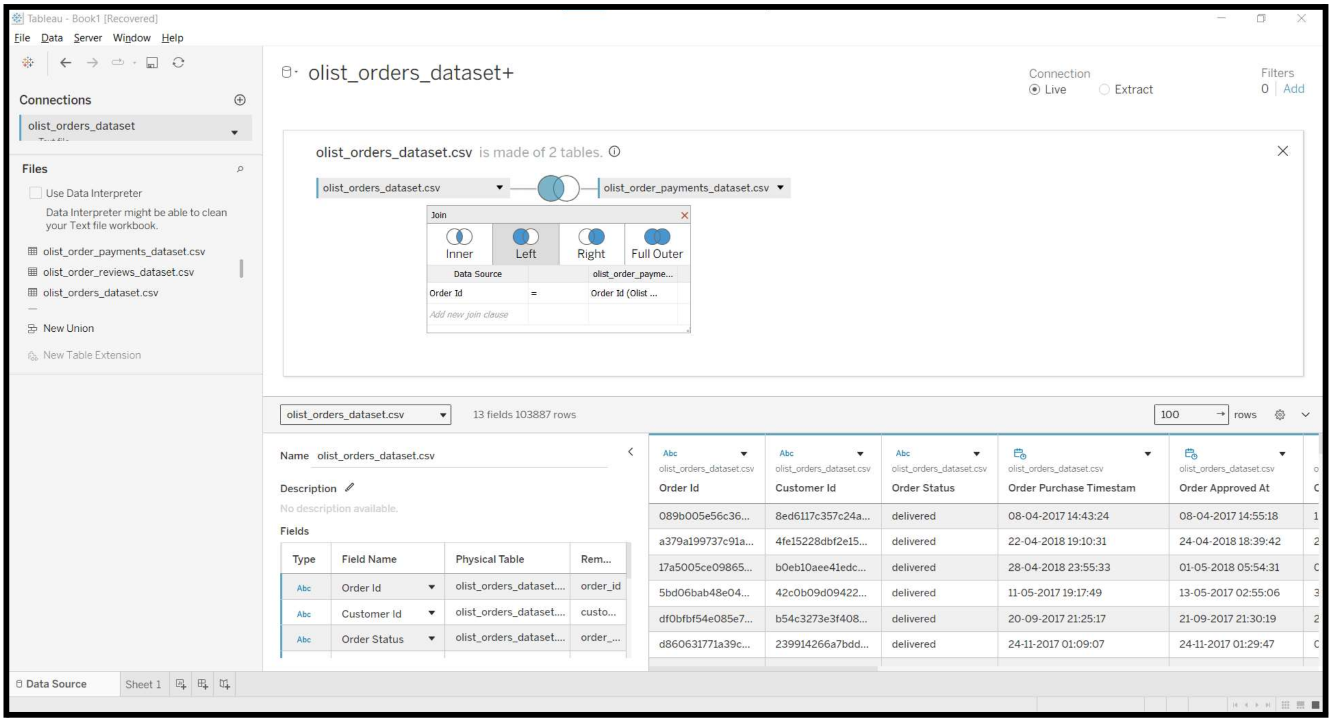
Screenshot 10: ***Physical Layer displaying the olist_orders_dataset.csv and olist_order_payments_dataset.csv tables.***



Screenshot 11: ***Join configuration with order_id selected as the matching field.***



Screenshot 12: ***Left Join successfully applied between the Orders and Order Payments tables.***

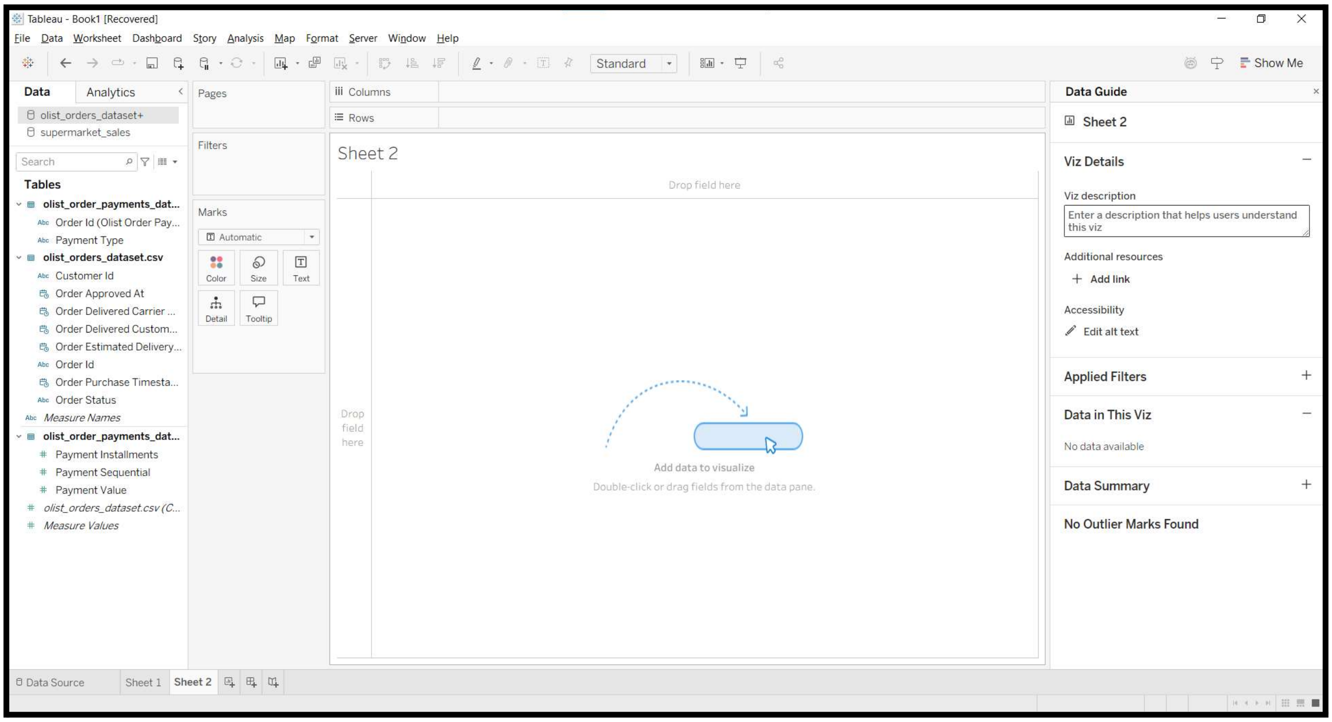


E. Applying Data Blending Between Independent Sources.

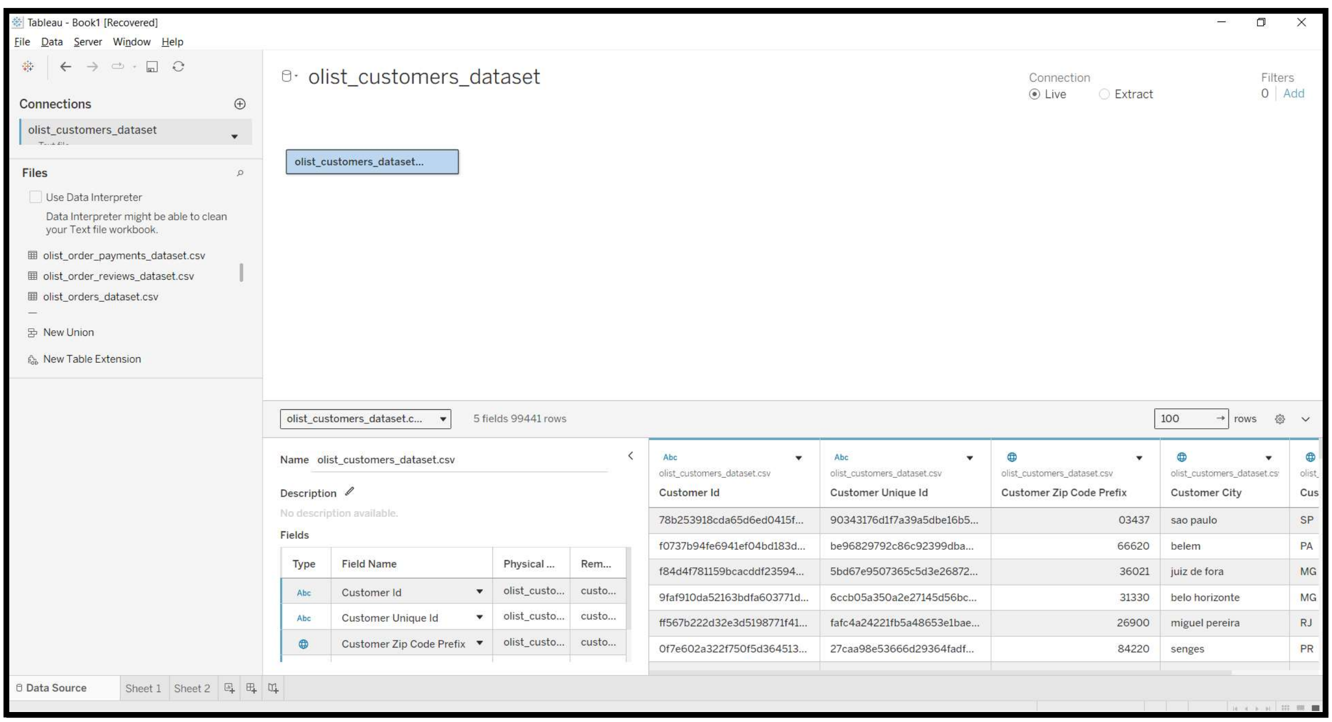
Step 5 – Apply Data Blending Between Independent Data Sources

1. Create a new worksheet by clicking the **New Worksheet (+)** icon to open **Sheet 2**.
2. In the **Data Pane**, select the **Orders** data source and drag the **Status** field to the **Rows** shelf.
3. Go to **Data** → **New Data Source** and add the **olist_customers_dataset.csv** file.
4. Return to **Sheet 2**, then click **Data** → **Edit Blend Relationships**.
5. Set **Orders** as the **Primary** data source and **Customers** as the **Secondary** data source.
6. Change the relationship from **Automatic** to **Custom**, then map **customer_id** from both datasets and click **OK**.
7. If **customer_id** is hidden, return to the **Data Source**, switch the connection to **Live**, go back to **Sheet 2**, select **Show Hidden Fields**, and unhide **customer_id**.
8. In the **Customers** data source, click the chain icon next to **customer_id** to activate the blend.
9. Drag **customer_state** from the **Customers** data source to the **Columns** shelf.

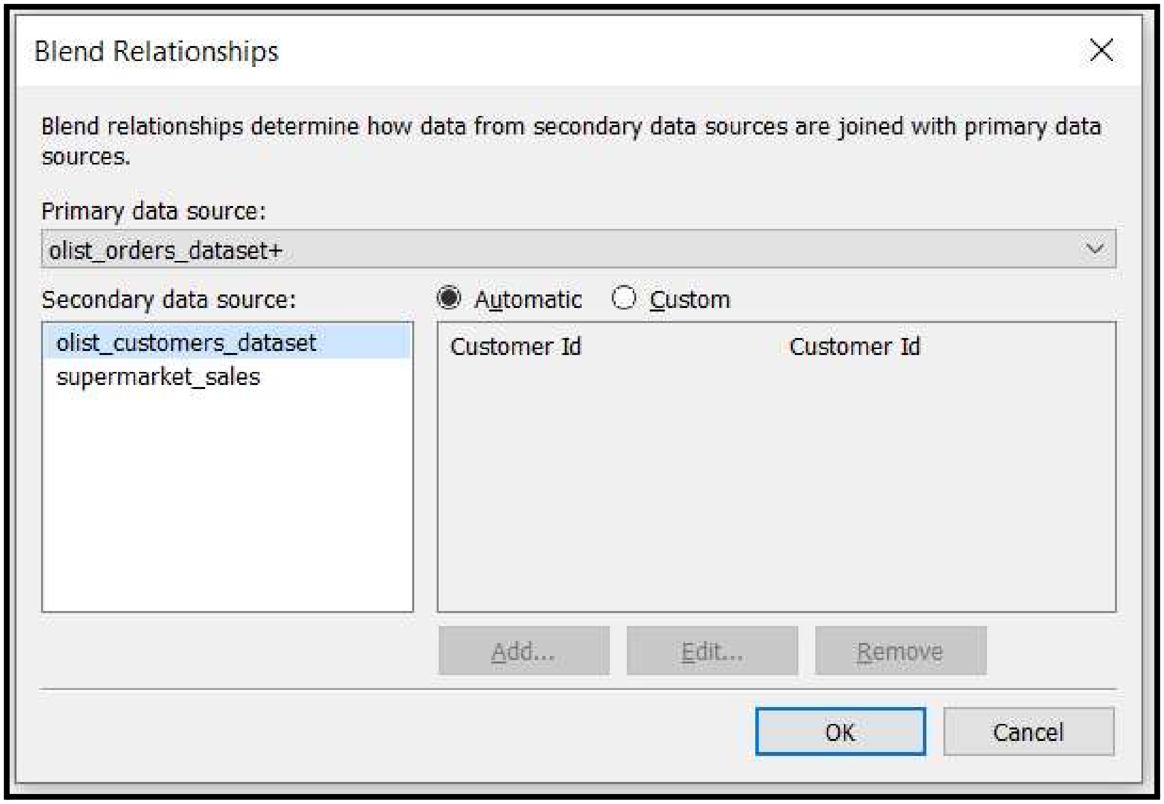
Screenshot 13: ***New worksheet (Sheet 2) created with the Orders data source selected.***



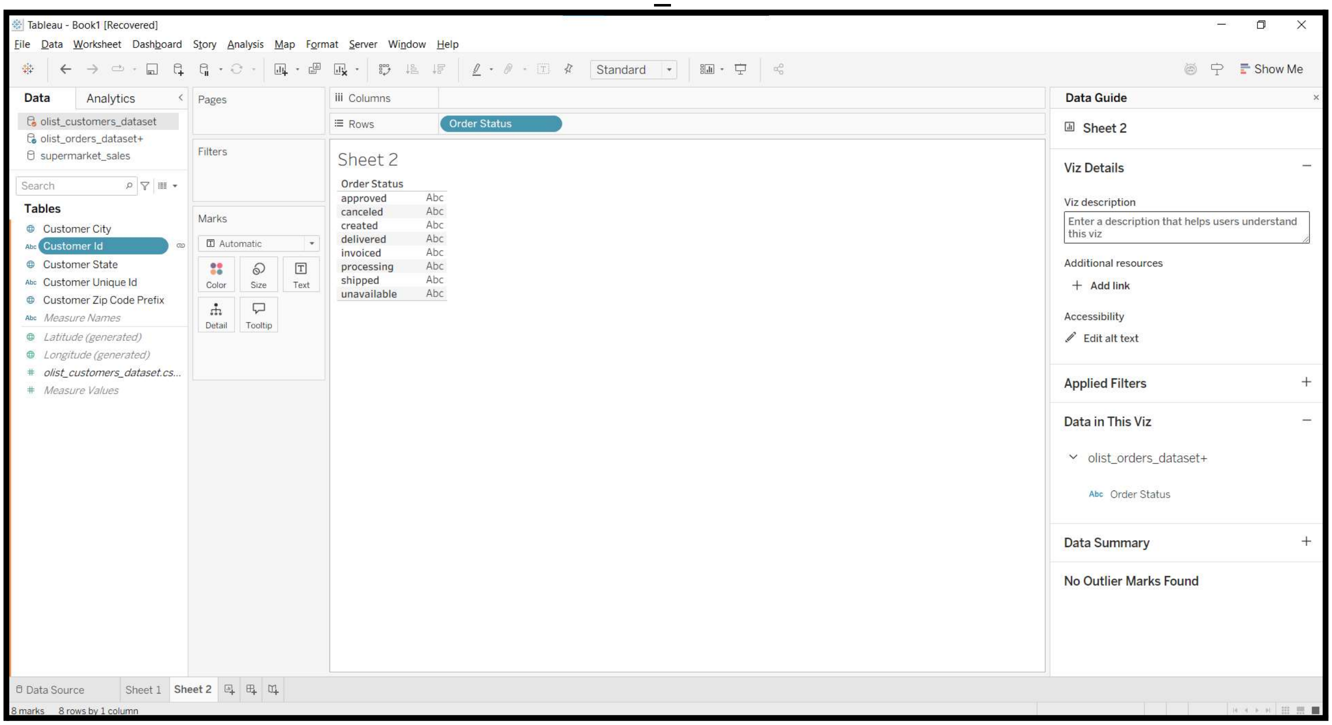
Screenshot 14: ***Customers dataset added as a new data source.***



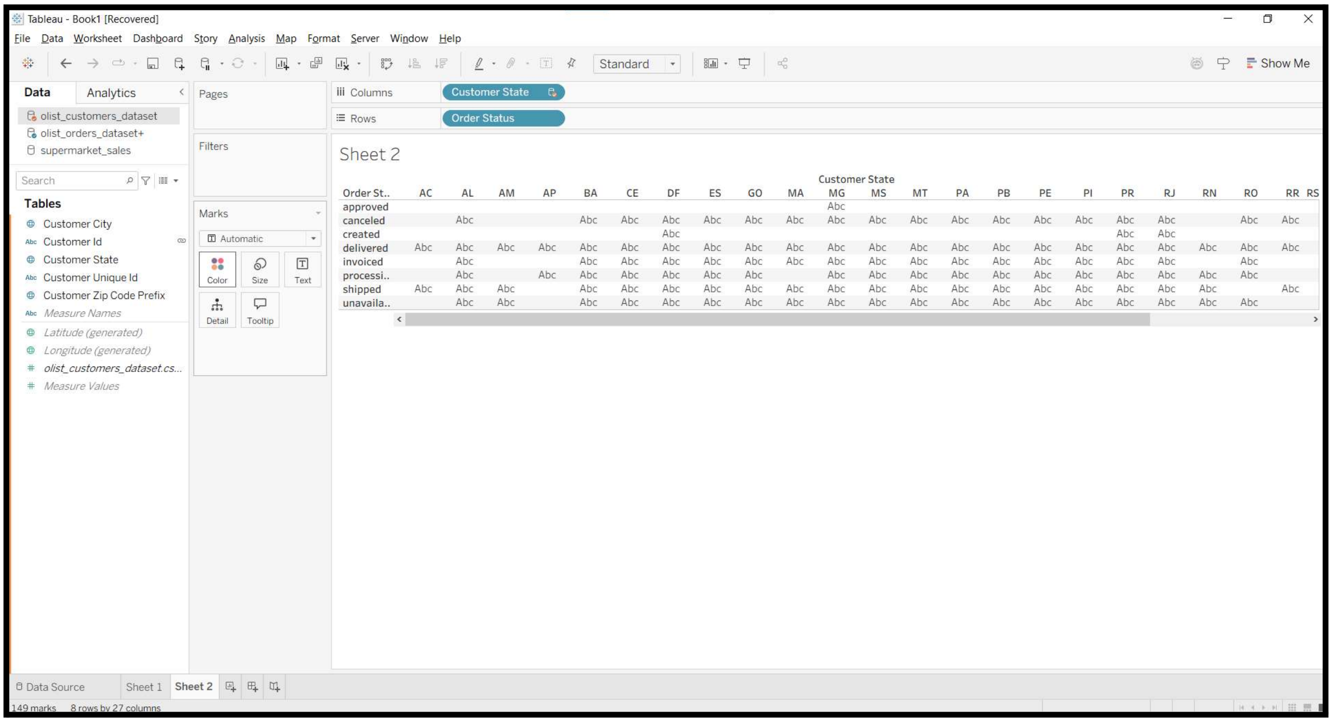
Screenshot 15: ***Edit Blend Relationships configured using customer_id.***



Screenshot 16: ***Chain link activated on customer_id to establish the data blend.***



Screenshot 17: ***Blended view displaying Status and customer_state on the worksheet.***

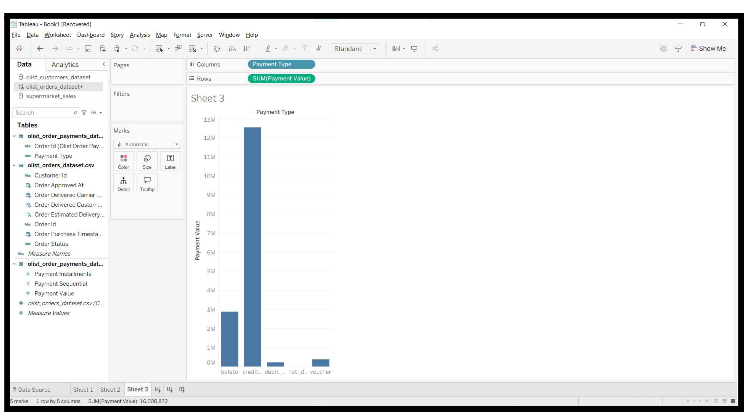


F. Implementing Multi-Tiered Filters.

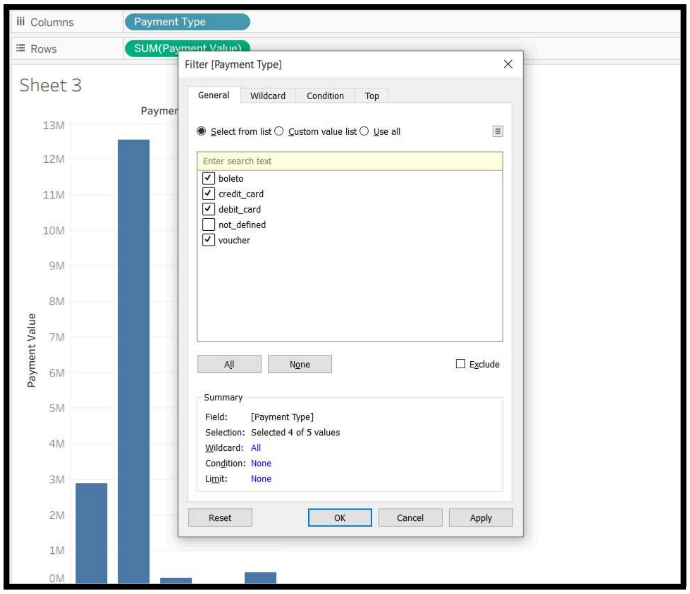
Step 6 – Apply Dimension, Measure, and Date Filters

1. Open a new worksheet and drag **Payment Value** to the **Rows** shelf and **Payment Type** to the **Columns** shelf.
2. Drag **Payment Type** to the **Filters** shelf, uncheck **not_defined**, select **credit_card** and **boleto**, then click **OK**.
3. Drag **Payment Value** to the **Filters** shelf, choose **All Values**, click **Next**, select **At Least**, enter **50**, and click **OK**.
4. Drag **Order Purchase Timestamp** to the **Filters** shelf, choose **Years** or **Relative Date**, select the required date range, and click **OK**.

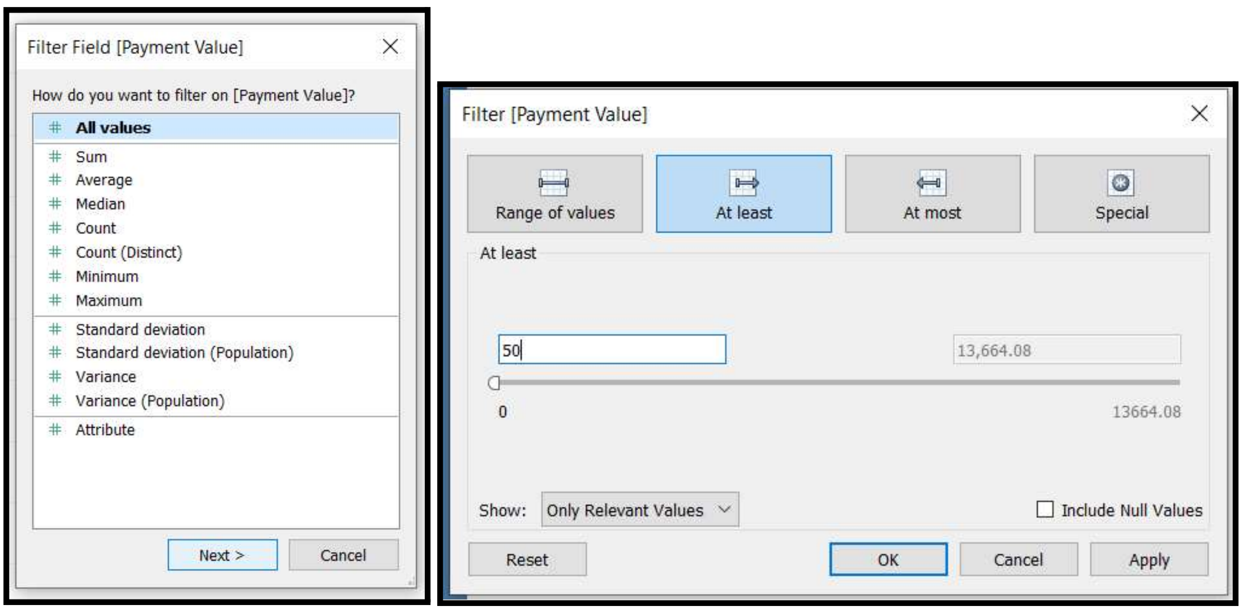
Screenshot 18: ***Payment Value and Payment Type added to the worksheet.***



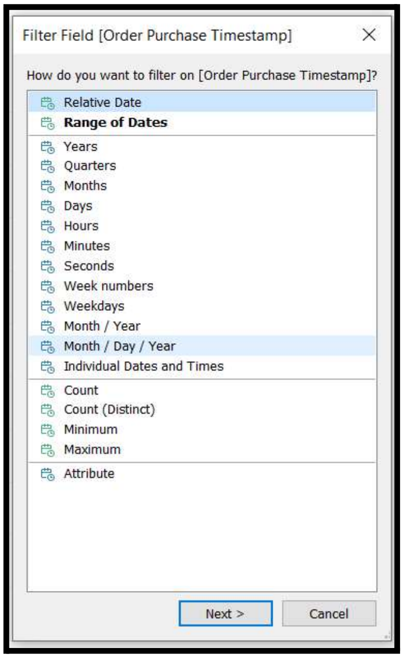
Screenshot 19: ***Payment Type filter applied with credit_card and boleto selected.***

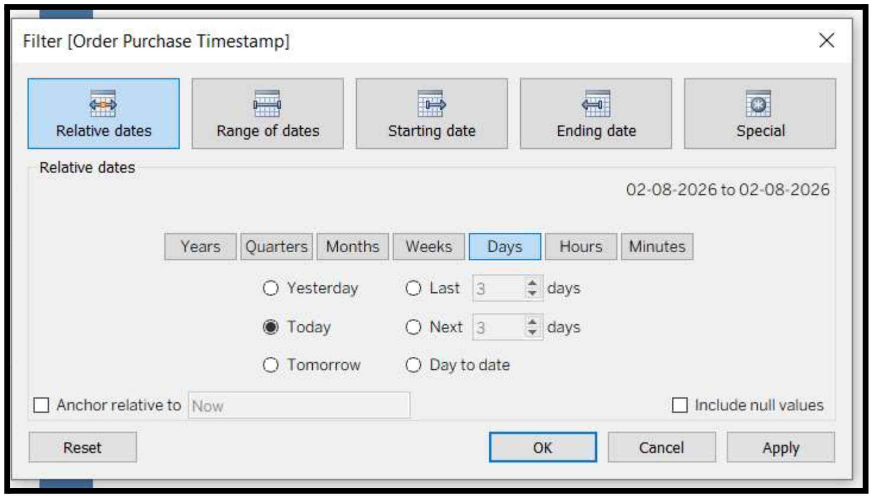


Screenshot 20: ***Payment Value filter set to show values greater than or equal to 50.***



Screenshot 21: ***Date filter applied using the Order Purchase Timestamp field.***



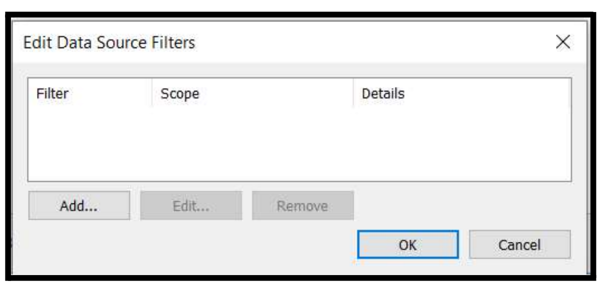


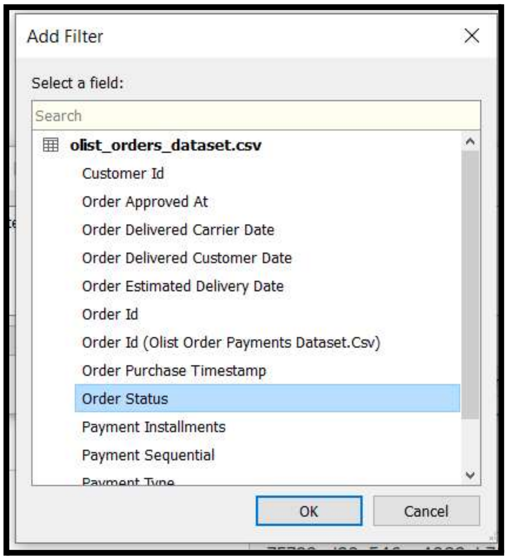
G. Optimizing Dashboard Performance.

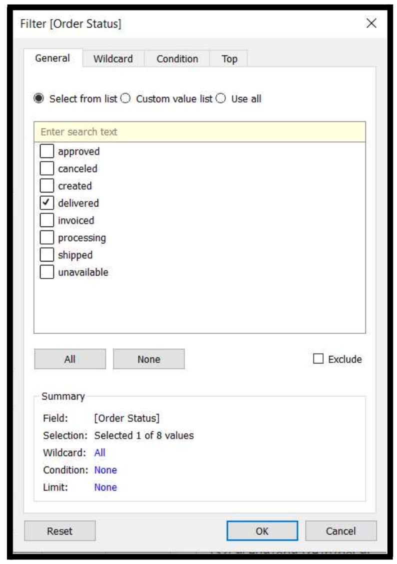
Step 7 – Apply Extract and Context Filters

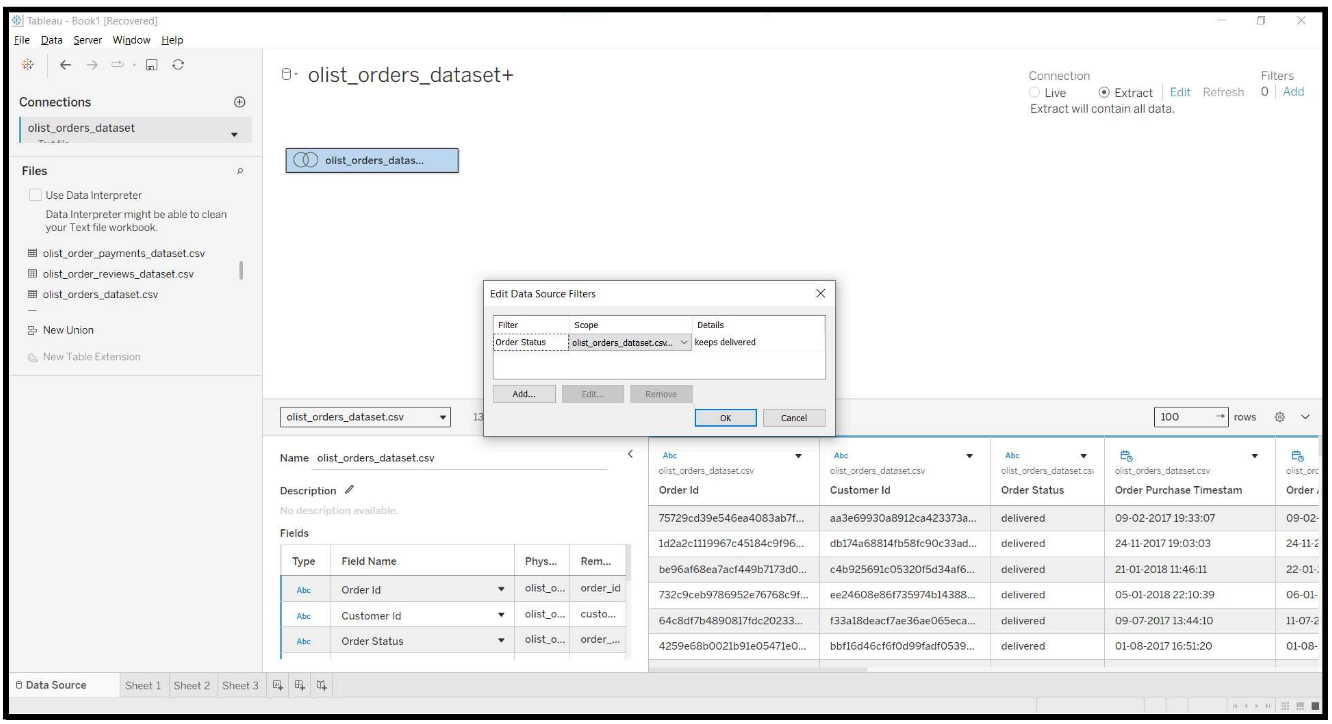
1. Go to the **Data Source** page and click **Edit** under the **Extract** option.
2. Click **Add**, select the **Order Status** field, choose **delivered**, and click **OK** to create an extract containing only delivered orders.
3. Return to the worksheet and locate the **Filters** shelf.
4. Right-click any active filter (such as **Payment Type** or **Order Status**) and select **Add to Context**. The filter will turn **gray**, indicating it has become a context filter.

Screenshot 22: ***Extract Filter configured to include only delivered orders.***









Screenshot 23: ***Context Filter applied, showing the filter pill in gray on the Filters shelf.***